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# Instruct-4DGS: Efficient Dynamic Scene Editing via 4D Gaussian-based Static-Dynamic Separation

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## Abstract

*Recent 4D dynamic scene editing methods require editing thousands of 2D images used for dynamic scene synthesis and updating the entire scene with additional training loops, resulting in several hours of processing to edit a single dynamic scene. Therefore, these methods are not scalable with respect to the temporal dimension of the dynamic scene (i.e., the number of timesteps). In this work, we propose* ***Instruct-4DGS****, an efficient dynamic scene editing method that is more scalable in terms of temporal dimension. To achieve computational efficiency, we leverage a 4D Gaussian representation that models a 4D dynamic scene by combining static 3D Gaussians with a Hexplane-based deformation field, which captures dynamic information. We then perform editing solely on the static 3D Gaussians, which is the minimal but sufficient component required for visual editing. To resolve the misalignment between the edited 3D Gaussians and the deformation field, which may arise from the editing process, we introduce a refinement stage using a score distillation mechanism. Extensive editing results demonstrate that Instruct-4DGS is efficient, reducing editing time by more than half compared to existing methods while achieving high-quality edits that better follow user instructions. Code and results:* https://hanbyelcho.info/instruct-4dgs/

## 1. Introduction

Diffusion-based generative models [10, 21, 42, 46, 48, 49, 68] have recently achieved remarkable progress in the 2D image domain and are increasingly being integrated into practical applications. As the demand for generative tasks extends beyond 2D, the editing of 3D and 4D dynamic scenes has emerged as a significant area of research. In particular, user-instruction-guided editing is gaining traction as an intuitive and user-friendly approach.

In this context, InstructPix2Pix (IP2P) [4] has gained recognition by proposing a novel method for editing 2D images based on user instructions. Building on IP2P's capabilities, research on instruction-guided 3D scene editing, particularly with NeRF [37] and 3D Gaussian Splatting (3DGS) [24], has become increasingly active. However, 4D dynamic scene editing remains relatively underexplored. One of the few existing methods, Instruct 4D-to-4D [38] requires iterative dataset updates for *"thousands of 2D images"* used in the dynamic scene synthesis, as shown in Fig. 1 (a), along with additional training loops to update the entire dynamic scene, resulting in several hours of processing to edit a single dynamic scene. Regardless of how efficiently the dataset is updated, such an approach fails to scale with the temporal dimension of dynamic scenes, making it impractical for real-world applications.

In this work, we propose **Instruct-4DGS**, an efficient 4D dynamic scene editing method that is more scalable with respect to the temporal dimension. To maximize computational efficiency, we focus on three key aspects: (1)

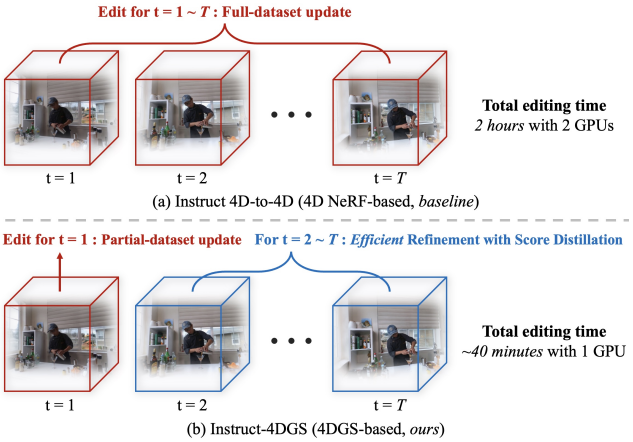


Figure 1. **Illustration of dynamic scene editing processes for baseline and our method**: (a) The existing method requires updating the 2D images for all timesteps. (b) In contrast, our method updates only the first timestep's dataset images, edits canonical 3D Gaussians, and efficiently completes dynamic scene editing through score-based temporal refinement. For a multi-camera dataset with $T = 300$, our method reduces editing time by more than half compared to the baseline, using only a single GPU.

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Since 4D dynamic scenes require frequent rendering during the editing process, we employ 4D Gaussian Splatting (4DGS) [63] as our scene representation, enabling fast and efficient rendering. (2) Our objective is to edit the appearance of the scene while preserving its motion. To achieve this, we leverage the inherent separability of 4DGS into static and dynamic components—specifically, canonical 3D Gaussians (*static*) and a Hexplane [5, 15]-based deformation field (*dynamic*)—allowing us to improve efficiency by editing only the static component. (3) To ensure better alignment between the edited static 3D Gaussians and the original deformation field, we perform temporal refinement using a score distillation mechanism [43].

Specifically, the Hexplane-based 4DGS offers notable advantages in both editing quality and rendering efficiency compared to the 4D NeRF [55] used in Instruct 4D-to-4D. By employing 3D Gaussians to represent the static canonical scene, we ensure high-quality, real-time rendering during the editing process. Additionally, Hexplane, which utilizes a spatio-temporal encoding structure based on planar factorization, is highly compact, further contributing to real-time rendering performance.

In addition to rendering efficiency, we aim to achieve computational efficiency in dynamic scene editing by focusing solely on the static component. Since our goal is to edit the scene's appearance while preserving its motion, we modify only the static 3D Gaussians, which are the minimal yet sufficient elements for appearance editing. As shown in Fig. 1 (b), this approach allows us to edit the entire dynamic scene without updating every 2D images, even for scenes with extended timesteps. Specifically, we edit only a subset of 2D multiview images from the initial timestep using IP2P and then apply simple modifications to the static 3D Gaussians using L1 RGB loss.

While editing only the static 3D Gaussians is simple and efficient, it introduces motion artifacts in later timesteps. Specifically, modifying the static 3D Gaussians causes slight shifts in the positions of Gaussian primitives, leading to misalignment between the static canonical scene and the original deformation field. Additionally, only the Spherical Harmonics (SH) colors of 3D Gaussians visible in the first timestep are updated. As a result, when Gaussian primitives rotate through the deformation field in subsequent timesteps, previously unmodified SH values become exposed, introducing visual artifacts. In summary, the dynamic scene tends to overfit to the first timestep, leading to artifacts across other timesteps.

To address this temporal misalignment, we propose a refinement stage that adjusts the edited static 3D Gaussians to better align with the original deformation fields. Specifically, we utilize the score distillation mechanism proposed in DreamFusion [43] to transfer IP2P's editing guidance into 3D and even 4D spaces. We apply a score-based refinement stage to eliminate artifacts in the *pseudo-edited dynamic scene*, where the edited static 3D Gaussians are misaligned with the deformation field. Additionally, inspired by MVDream [53] and Tune-a-Video [64], we replace IP2P's self-attention module with a cross-attention module. This modified IP2P, Coherent-IP2P prevents the accumulation of non-uniform editing guidance during score distillation, which would otherwise result in blurry outputs.

Our evaluation demonstrates a significant reduction in editing turnaround time while improving visual quality. Furthermore, our method can effectively perform dynamic scene editing across various user instructions. Our main contributions are summarized as follows:

- We propose Instruct-4DGS, the first efficient dynamic scene editing framework based on 4D Gaussian Splatting.
- We achieve efficient dynamic scene editing by modifying only static 3D Guassians, the minimal but sufficient component for visual editing.
- We propose a refinement method using score distillation with Coherent-IP2P, which removes motion artifacts while maintaining computational efficiency.
- Our method reduces editing time by more than half while achieving higher visual quality.

## 2. Related Work

### 2.1. 4D Dynamic Scene Representation

Recent advancements in computer vision and graphics have fueled interest in 4D dynamic scene representation, which models both spatial and temporal information. As high-quality 4D content capture continues to improve, multi-view 4D data has become increasingly available, highlighting the need for efficient representations to mitigate the high computational costs of 4D modeling. Many approaches [12, 34, 40, 44, 51, 55, 59] have reduced the complexity of dynamic scene representation by handling the temporal dimension separately, leading to the decoupling of the canonical 3D representation and the deformation field. Specifically, K-plane and Hexplane [5, 15] construct spatio-temporal encoding structures within the deformation field using multi-scale parameter grids through planar factorization. Other methods [13, 30, 31, 66] have enhanced the overall performance of dynamic scenes by employing 3D Gaussian Splatting [24] as the canonical 3D representation, which has recently gained attention for its real-time rendering capabilities and high visual quality. Notably, 4D Gaussian Splatting [63] combines 3DGS with the Hexplane deformation field to achieve real-time rendering speeds while more accurately modeling dynamic scenes. Given its excellent performance, 4DGS holds great potential for dynamic scene generation [1, 33, 47], editing [38, 52], and tracking [17, 36]. In this paper, we employ 4DGS to maximize the efficiency of the dynamic scene editing process.

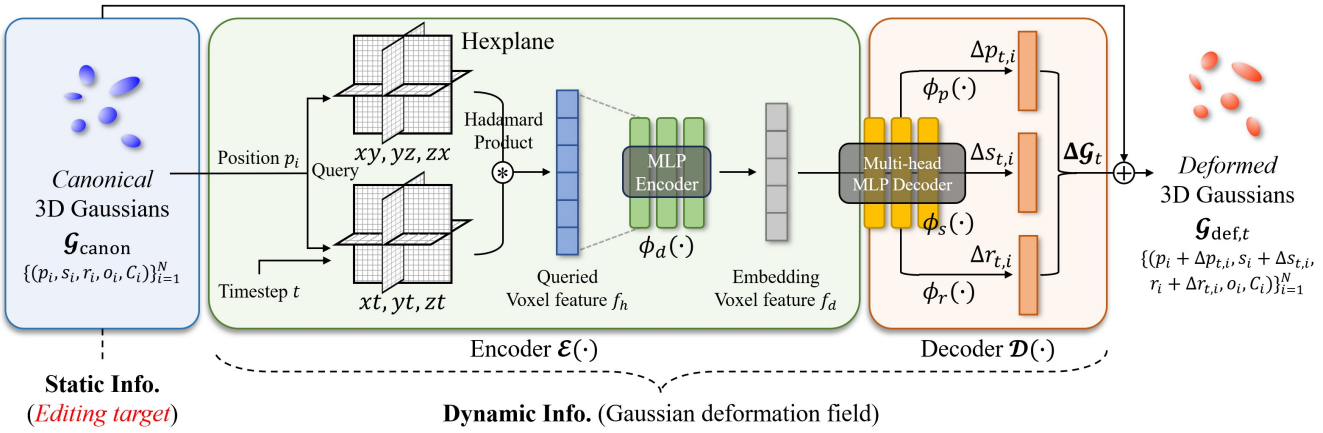


Figure 2. **Overview of 4D Gaussian Splatting**: 4DGS represents dynamic scenes by separating static (canonical 3D Gaussians $\mathcal{G}_{\text{canon}}$) and dynamic components (Gaussian deformation field $\{\mathcal{E}(\cdot), \mathcal{D}(\cdot)\}$). Given a Gaussian primitive's position $p$ and timestep $t$, a spatio-temporal embedding voxel feature $f_d$ is queried from the Hexplane. This feature is then processed by a multi-head MLP decoder $\{\phi_p(\cdot), \phi_s(\cdot), \phi_r(\cdot)\}$ to generate per-Gaussian deformation parameters $\Delta p_{t,i}$, $\Delta s_{t,i}$, and $\Delta r_{t,i}$. By adding these parameters to the canonical Gaussians $\mathcal{G}_{\text{canon}}$, we obtain the deformed Gaussians $\mathcal{G}_{\text{def},t}$. Finally, by repeatedly generating and rendering $\mathcal{G}_{\text{def},t}$ across timesteps, the dynamic scene video is obtained.

### 2.2. Instruction-Guided Scene Editing

User instructions provide one of the most intuitive and user-friendly approaches to scene editing. InstructPix2Pix [4] introduced instruction-guided editing by fine-tuning the Stable Diffusion [48] model on a dataset of *source image–instruction–target image* triplets. Recent studies [7, 8, 11, 18, 23, 65] have extended IP2P's capabilities to 3D scenes by developing methods to ensure spatial consistency in editing guidance, thereby making significant progress in instruction-guided 3D scene editing despite the limited availability of 3D datasets. Among these approaches, one of the key trends is the iterative dataset update method, where all 2D images used for 3D scene synthesis are edited, followed by re-training the 3D scene. Recently, Instruct 4D-to-4D [38] extended this iterative dataset update approach to 4D space, presenting the first instruction-guided 4D editing method. By employing flow-based [58] and depth-based warping to ensure spatio-temporal consistency during dataset updates, they achieved notable results. However, editing all images for 4D dynamic scenes remains extremely time-consuming, highlighting the need for a more efficient approach that can effectively leverage diffusion priors for dynamic scene editing. Therefore, we propose an efficient dynamic scene editing method that significantly reduces total editing time.

### 2.3. Score Distillation Sampling

The Score Distillation Sampling (SDS) mechanism was introduced in DreamFusion [43] for text-to-3D scene generation, enabling the transfer of pre-trained 2D diffusion model priors [4, 21, 48] to other data domains. When SDS is used with diffusion networks incorporating specific hypotheses—such as multiview diffusion models [53, 60] or video diffusion models [2, 3, 22, 35, 54, 64]—it produces guidance that aligns with those hypotheses. Many studies [1, 32, 33, 43, 47, 57, 67] have leveraged this property to develop SDS-based 3D/4D generation methods. Meanwhile, other approaches [9, 23, 29, 62, 65, 70] have explored SDS for editing tasks, adapting it to improve spatial and temporal consistency during the editing process. Furthermore, some works [19, 25, 27] have enhanced editing performance by modifying the score loss function to better suit editing-specific objectives.

## 3. Preliminary

For efficient dynamic scene editing, we leverage 4D Gaussian Splatting (4DGS) [63] which represents scenes by separating static and dynamic information. In this section, we briefly review 4DGS, highlighting its Hexplane [5, 15]-based Gaussian deformation field, and introduce our proposed method in Sec. 4.

**4D Gaussian Splatting.** 4DGS consists of a canonical 3D Gaussians [24] $\mathcal{G}_{\text{canon}}$ that represents static information and a Gaussian deformation field that represents dynamic information and produces each Gaussian's deformation $\Delta\mathcal{G}_t$ (where $t$ is a normalized value from 0 to 1, denoting the timestep within the dynamic scene), as illustrated in Fig. 2. The 3D Gaussians $\mathcal{G}_{\text{canon}}$ representing the undeformed static canonical 3D scene consist of $N$ Gaussian primitives (in our case, $N = 100k$–$200k$), denoted as $\mathcal{G}_{\text{canon}} = \{(p_i, s_i, r_i, o_i, C_i)\}_{i=1}^N$, where each primitive is defined by a position $p \in \mathbb{R}^3$, a scaling vector $s \in \mathbb{R}^3$, a rotation quaternion $r \in \mathbb{R}^4$, an opacity $o \in \mathbb{R}$, and a spherical harmonics color $C \in \mathbb{R}^k$, with $k$ determined by the SH